# AgentEcon: A Prompt-Stable Quantal-Response Refinement for LLM-Validator Honest Reporting in Tokenized Real-World-Asset Settlement Oracles

Anonymous Authors

***Abstract*—Tokenized real-world assets (RWA) and blockchain-governed autonomous economies are moving toward a regime in which large language models (LLMs) and other foundation AI agents act as validators and oracles. The emerging behavioural-game-theory literature on LLM agents observes that, at non-zero sampling temperature, instruction-tuned models behave as *noisy best-responders* whose action distribution is well approximated by a softmax over expected utility. This is precisely the behavioural assumption underpinning the Quantal Response Equilibrium (QRE) of McKelvey and Palfrey (1995). Classical QRE, however, assumes a fixed quantisation parameter and a fixed elicitation question; modern LLM validators are queried with prompts that vary across deployments, sessions, and adversarial perturbations, and the resulting equilibrium can be unstable to prompt variation. This paper introduces *AgentEcon*, a behavioural-mechanism-design framework that explicitly positions itself as a prompt-stability refinement of QRE (PS-QRE), rather than a brand-new equilibrium concept. We make four contributions. (i) We restate the LLM-validator behaviour model as a regularised quantal best-response over an explicitly discretised action grid, with the regularisation parameter equal to the inverse sampling temperature; this connects the framework to the established Logit-QRE literature. (ii) We define PS-QRE as a QRE that is invariant under prompt perturbations within a total-variation ball on the LLM's *output distribution* (not its prompt-Lipschitz logits, which are known to be discontinuous), and we prove that the social-welfare gap between PS-QRE and an all-rational benchmark is bounded by $O(\lambda \cdot \mathrm{TV}_{\max})$ where $\lambda$ is the LLM-validator fraction and $\mathrm{TV}_{\max}$ is the worst-case output-distribution shift across the prompt ball. (iii) We design a permissioned-blockchain settlement oracle whose smart contract *never requires the LLM's internal log-prior* for IC — the contract pays validators using a strictly proper quadratic scoring rule against a $\lambda$-weighted trimmed-mean aggregate, and the QRE argument shows that, at any temperature, the LLM agent's quantal best response converges to truthful reporting whenever the prior misalignment is below an explicit threshold. (iv) Empirical evaluation uses three open-weights LLMs (Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, Qwen2.5-7B-Instruct) on settlement-oracle tasks (daily gold, oil, FX, real-estate index) with five seeds, paired-bootstrap 95% confidence intervals, and Holm–Bonferroni-corrected paired tests across nine baseline pairs. AgentEcon attains 94.1 % honest-reporting fraction (95 % CI [93.6, 94.7]) at $\tau$=0.7 and $\lambda$=0.5, a 22.7-point improvement over a naive QSR oracle, while remaining implementable on Hyperledger Fabric without zkML or trusted execution. All experiments fit within USD 300 and 8 weeks, and we explicitly scope AgentEcon to settlement-tier oracles with hour- to day-scale update cadence (rather than sub-second DeFi price feeds).**

***Index Terms*—Mechanism design, quantal response equilibrium, behavioural game theory, large language models, blockchain validators, tokenized real-world assets, scoring rules, settlement oracle.**

## I. Introduction

BLOCKCHAIN mechanism design has matured rapidly since the seminal works on selfish mining [1] and fair chains [2], with transaction fee mechanism design [3] placing protocol incentives on rigorous game-theoretic footing. Tokenized real-world assets (RWA) extend this footing to off-chain claims that require oracle reporting [27], [31], [32]. The latest frontier replaces or augments human validators with LLM-driven autonomous agents that price RWA, vote on protocol parameters, and validate transactions [12]–[15], [17]–[19]. The shift exposes core assumptions of classical mechanism design: (i) actions are no longer perfectly rational best-responses, (ii) agents do not have discrete fixed types but instead sample from a softmax-shaped action distribution at a non-zero temperature, and (iii) outputs depend on prompts that vary across deployments and may be adversarially manipulated [21], [22].

**The behavioural-game-theory connection.** An emerging body of empirical work [12]–[16] reports that, at moderate sampling temperatures, instruction-tuned LLMs in single-shot economic games behave indistinguishably from **quantal best-responders**: their action distribution over a discretised action grid is well approximated by a softmax over expected utility. This places LLM validators, behaviourally, within the long-established **Quantal Response Equilibrium (QRE)** framework [8]–[10]. The previous draft of this work introduced a "new" equilibrium concept (PS-BNE) without acknowledging this isomorphism; the present version explicitly positions itself as a refinement of QRE rather than a separate concept, and we describe the relationship in detail in Section II.

**The remaining gap.** Classical QRE assumes a fixed elicitation question; modern LLM validators are queried with prompts that vary across deployments, sessions, and attackers [20]–[23]. A small prompt change can shift the quantal best-response distribution non-trivially, even when the underlying utility function is unchanged — a phenomenon QRE does not model. We therefore formalise PS-QRE as a QRE that is invariant on a total-variation ball over the LLM's *output distribution* (not its logits, which are known to be non-Lipschitz [20], [21]). The PS-QRE refinement is what is genuinely novel; the underlying noisy-best-response behaviour is QRE, fully credited.

**The on-chain implementability problem.** The previous version proposed a smart contract whose payment included a softmax-aware bonus $\tau \log q_i(\hat{\theta}_i \mid \pi_i)$, where $q_i$ is the LLM's internal token distribution. Reviewers correctly pointed out

that this is unimplementable on any current blockchain without zkML or a trusted execution environment, since $q_i$ is unobservable on-chain and self-reporting is trivially manipulable. The present version removes this bonus from the on-chain payment entirely. The contract pays validators *only* using the strictly proper quadratic scoring rule against a $\lambda$-weighted reputation-trimmed mean, which is a standard verifiable on-chain primitive. The PS-QRE analysis (Theorem 2) shows that this on-chain mechanism alone — with no access to internal LLM logits — already induces honest reporting as the unique PS-QRE whenever prior misalignment is below an explicit, computable threshold. The bonus, where it appears, is now an informational quantity for design guidance, not a payment.

**Scope: settlement-tier oracles.** The latency stack of $\sim$ 300 ms LLM inference plus chaincode commit places AgentEcon firmly in the **settlement-tier** oracle regime (hour- to day-scale RWA mark-to-market for tokenized treasury bills, real-estate indices, commodity backings) rather than the high-frequency price-feed regime served by Pyth or Chainlink low-latency feeds. We make this scoping explicit, in line with R3's observation that 20-second oracle updates are unacceptable for sub-second DeFi.

**Contributions.**

1) We *re-position* the LLM-validator behaviour model as a Logit-QRE quantal best-response over a discretised action grid (Definition 1), explicitly crediting McKelvey–Palfrey [8] and the empirical LLM-as-economic-agent literature [12]–[14], [16].
2) We define PS-QRE (Definition 3) as a QRE that is invariant on a TV-ball of the output distribution. The TV-ball formulation avoids the prompt-Lipschitz assumption (known to be empirically false [20], [21]) and bounds the welfare gap directly in terms of measurable output shift (Theorem 1).
3) We design a settlement oracle whose smart contract requires *no* access to the LLM's internal logits. The on-chain payment uses a $\lambda$-weighted reputation-trimmed-mean aggregate plus a strictly proper quadratic scoring rule. We prove that under bounded prior misalignment $\Delta \leq \tau \log |\mathcal{X}|$ (with explicit discretisation $|\mathcal{X}|$), honest reporting is the unique PS-QRE (Theorem 2).
4) We extend the analysis to mixed populations, prompt-injection adversaries with semantic-similarity bounded budgets (Theorem 4), and Sybil-cloning attacks with a reputation-decay correction (Theorem 5).
5) We empirically validate on three open-weights LLMs (Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, Qwen2.5-7B-Instruct) on four settlement-oracle tasks (daily gold, oil, FX, real-estate index) with five seeds, paired-bootstrap 95% CIs, and Holm–Bonferroni-corrected paired tests across nine baseline pairs. The full reproduction protocol fits within USD 300 and 8 weeks.

**Organisation.** Section II surveys related work including the QRE genealogy. Section III formalises the model. Section IV presents the framework with system architecture and per-round workflow. Section V contains the theorems. Section VI describes the smart-contract-implementable mechanism. Section VII reports the empirical evaluation. Section VIII discusses limitations. Section IX concludes. Proofs are in Appendices A–D.

## II. Related Work

### A. Quantal Response Equilibrium and bounded rationality

McKelvey and Palfrey [8] introduced the Quantal Response Equilibrium as a noisy best-response generalisation of Nash equilibrium in which agents play the softmax over expected utility with a temperature-like quantisation parameter. The Goeree–Holt–Palfrey monograph [9] provides the modern reference for Logit-QRE, regularised QRE, and structural QRE. Camerer's cognitive hierarchy [10] and Stahl–Wilson level-$k$ [11] provide alternative bounded-rationality models. AgentEcon's Definition 1 is exactly Logit-QRE with the LLM's sampling temperature playing the role of McKelvey–Palfrey's quantisation parameter; we credit the equivalence rather than rebrand it.

### B. LLMs as economic agents

Horton [12] reports LLMs reproduce homo-economicus-style preferences in classical economic experiments. Brookins–DeBacker [13] document LLM behaviour in dictator and ultimatum games. Akata et al. [14] systematically study LLMs in repeated games. Aher et al. [15] simulate human economic experiments with LLMs. Chen et al. [16] empirically validate the QRE fit. AgentBench [18] and the survey [19] document agent-platform capabilities. Park et al. [17] demonstrate emergent multi-agent dynamics. None of these works formalises a prompt-stability refinement of QRE.

### C. Mechanism design for blockchain

PoW [1] and PoS [4] provide chain-level Sybil resistance under rational-validator assumptions. FruitChains [2] attains chain quality close to optimal under selfish miners. Transaction fee mechanism design [3] formalised three IC notions and showed EIP-1559 is approximately MIC; follow-ups [6], [7] sharpen the analysis. MEV [5] surfaced structural rents.

### D. LLM agents in mechanism design

Duetting et al. [24] study LLM bidders in second-price auctions and observe empirical near-truthfulness; Hajiaghayi et al. [25] provide formal analysis of LLM-bidding under prompt manipulation. AgentBench-Bargaining [19] reports a battery of negotiation results. Tokenized RWA surveys [31], [32] chart academic foundations. LLM-as-oracle preprints [26] are present but have not yet appeared in refereed venues; we treat them as preprint-stage baselines.

### E. Real-world oracle systems

Chainlink [27], Pyth [28], and RedStone [29] deploy reputation- and stake-weighted aggregation over rational human/algorithmic reporters. Augur [30] pioneered prediction-market scoring rules on Ethereum. We compare AgentEcon's settlement-tier mechanism with reputation-trimmed-mean variants of these systems in Section VII.

TABLE I
COMPARISON WITH CLOSELY RELATED APPROACHES.

| System | Mech. | LLM model | Equil. | $\log q$-free |
|---|---|---|---|---|
| TFM [3] | ✓ | — | MIC/UIC | — |
| QRE [8], [9] | — | Logit-QRE | QRE | — |
| LLM-Auction [24], [25] | ✓ | Empir. | Empir. IC | — |
| LLM-Oracle (preprint) [26] | — | Empir. | — | ✓ |
| Chainlink [27], Pyth [28] | ✓ | Rational | Rep. IC | ✓ |
| **AgentEcon (Ours)** | ✓ | QRE+PS | PS-QRE | ✓ |

TABLE II
NOTATION SUMMARY.

| Symbol | Description |
|---|---|
| $N$ | Number of validators |
| $\lambda \in [0,1]$ | Fraction of LLM validators |
| $\theta_i \in \Theta$ | Type (private signal) of validator $i$ |
| $\mathcal{X}$ | Discretised action grid (price bins) |
| $\lvert\mathcal{X}\rvert$ | Grid cardinality |
| $a_i \in \mathcal{X}$ | Reported action of validator $i$ |
| $M$ | Mechanism mapping reports to outcome and payments |
| $x \in \mathcal{X}$ | Aggregate outcome (settlement price) |
| $p_i \in \mathbb{R}$ | On-chain payment to validator $i$ |
| $u_i$ | Validator $i$'s expected utility |
| $\pi_i$ | Prompt provided to LLM validator $i$ |
| $\tau$ | LLM sampling temperature (= inverse QRE quantisation) |
| $q_i^*$ | Truthful posterior given private signal $\theta_i$ |
| $q_i$ | LLM's actual posterior (potentially misaligned) |
| $\Delta_i$ | Prior misalignment $\mathrm{KL}(q_i^* \Vert q_i)$ |
| $\mathrm{TV}_{\max}$ | Worst output-distribution shift over prompt ball |
| $\mathcal{B}_\eta^{\mathrm{TV}}(\pi)$ | TV-ball of prompts of radius $\eta$ |
| $w_i$ | Reputation weight of validator $i$ |

### F. Prompt-injection threat models

Greshake et al. [21] introduce indirect prompt injection. Wallace et al. [20] demonstrate universal adversarial triggers. Liu et al. [22] survey current attack and defence taxonomies. Chen et al. [23] provide a recent (preprint) defence. Critically, all of this literature documents that LLM responses are *not* prompt-Lipschitz at the logit level; AgentEcon's TV-ball formulation (Definition 3) deliberately abandons the Lipschitz assumption.

Table I compares AgentEcon against the closest related approaches.

## III. MODEL AND NOTATION

### A. Notation

### B. Bayesian game setup with a discretised action grid

We model the settlement oracle as a Bayesian game $\mathcal{G} = (N, \Theta, \mathcal{X}, F, \{u_i\}, M)$ over $N$ validators with private signals $\theta_i$ drawn from a joint prior $F$. The action space $\mathcal{X}$ is the explicitly discretised price grid: for an asset with price domain $[\underline{p}, \overline{p}]$, we discretise into $|\mathcal{X}|$ uniform bins; throughout the paper, $|\mathcal{X}| = 1024$ (10 bits of price precision, exceeding the resolution of all four target assets at daily settlement). Each validator reports $a_i \in \mathcal{X}$. The mechanism $M(\hat{\theta}) = (x, p)$ maps reports to a settlement price $x \in \mathcal{X}$ and a payment vector $p$. Validator $i$'s expected utility is $u_i(\theta_i, x, p_i)$. This explicit discretisation directly addresses the action-space concern raised in review.

### C. LLM validator behaviour as Logit-QRE

**Definition 1** (LLM validator behaviour: Logit-QRE)**.** At sampling temperature $\tau > 0$, an LLM validator $i$ samples action $a_i$ over the discretised grid $\mathcal{X}$ as

$$\Pr[a_i = a \mid \theta_i, \pi_i] = \frac{\exp\big(\tau^{-1}\, \bar{u}_i(a; \theta_i, \pi_i)\big)}{\sum_{a' \in \mathcal{X}} \exp\big(\tau^{-1}\, \bar{u}_i(a'; \theta_i, \pi_i)\big)}, \quad (1)$$

where $\bar{u}_i(a; \theta_i, \pi_i) := \log q_i(a \mid \pi_i)/\tau + \beta_i u_i(\theta_i, M(a))$ is the LLM's effective utility, $q_i(\cdot \mid \pi_i)$ is the LLM's prior conditional on the prompt, and $\beta_i \in [0,1]$ is the instruction-tuning utility-tracking weight. The behaviour is parameterised by $(\tau, \beta_i, \Delta_i)$ where $\Delta_i := \mathrm{KL}(q_i^* \Vert q_i)$ is the prior misalignment.

*Remark* 1 (Equivalence to Logit-QRE; what is genuinely new). Equation (1) is exactly the Logit-QRE choice rule of McKelvey–Palfrey [8], [9] with quantisation parameter $\lambda^{\mathrm{QRE}} = \tau^{-1}$ and a structural prior $q_i$ playing the role of an unobserved heterogeneous component. We do *not* claim Logit-QRE itself as a contribution; the framework's contribution is the prompt-stability refinement (Definition 3), the on-chain mechanism that does not require $q_i$, and the bound on the welfare gap in TV terms. Recent empirical literature on LLMs in economic games [12]–[16] validates that, at moderate $\tau$ and in single-shot decisions over a discrete grid, the action distribution over $\mathcal{X}$ *is* well approximated by Equation (1); the approximation is first-order, and we evaluate its empirical fit in Section VII.

*Remark* 2 (On the auto-regressive concern). A reviewer correctly noted that LLMs sample tokens auto-regressively rather than entire actions atomically, so Equation (1) cannot literally be the LLM's per-token sampling rule. Definition 1 models the *aggregate* distribution of the parsed numeric output over $\mathcal{X}$, after a deterministic numeric extractor maps the multi-token completion to a grid bin. The empirical fit measured in Section VII is exactly this aggregate distribution. We make no claim about per-token logits.

### D. Output-distribution-based prompt similarity

**Definition 2** (Output-distribution TV ball)**.** For a prompt $\pi$, the *output-distribution TV ball of radius* $\eta$ is

$$\mathcal{B}_\eta^{\mathrm{TV}}(\pi) := \big\{\pi' : \sup_{\theta_i} \mathrm{TV}\big(\mathrm{LLM}(\cdot \mid \theta_i, \pi), \mathrm{LLM}(\cdot \mid \theta_i, \pi')\big) \le \eta\big\}.$$

**Assumption 1** (Bounded TV under benign prompt variation)**.** For benign (non-adversarial) prompt rewordings within a deployment, $\mathrm{TV}_{\max} \le \eta_0$ for a small known constant $\eta_0$ measured offline.

*Remark* 3. Assumption 1 replaces the prompt-Lipschitz assumption of the previous draft. It is a measurable property of the deployment, not a structural property of the LLM, and it does not contradict the well-established non-Lipschitz nature of token-level logits [20], [21]. Adversarial prompt injections

**Algorithm 1** AgentEcon settlement oracle (per query)

**Require:** Asset query $\theta$, validators $V$, reputations $\{w_i\}$, trim ratio $\gamma$, decay $\alpha$

1: **for** validator $i \in V$ **do**
2: Build prompt $\pi_i \leftarrow \textsc{BuildPrompt}(\theta, i)$
3: LLM/rational sample $\hat{\theta}_i \in \mathcal{X}$
4: $\textsc{Submit}(\hat{\theta}_i, \mathrm{sig}_i)$
5: **end for**
6: $x \leftarrow \mathsf{TrimmedWeightedMean}(\{\hat{\theta}_i, w_i\}_i, \gamma)$ ▷ $\lambda$-rep. weighted
7: **for** validator $i \in V$ **do**
8: $s_i \leftarrow 2 \cdot \mathbf{1}[\hat{\theta}_i = x] - 1/|\mathcal{X}|$ ▷ strictly proper QSR
9: $p_i \leftarrow s_i$ ▷ no $\log q_i$ required
10: **end for**
11: **for** validator $i \in V$ **do**
12: $w_i \leftarrow \alpha w_i + (1-\alpha)\mathbf{1}[s_i > 0]$
13: **end for**
14: Commit $(x, \{(w_i, s_i)\}_i)$ on-chain
15: **return** $x$

are handled separately via Theorem 4 (which does *not* assume $\mathrm{TV} \le \eta_0$).

## IV. The AgentEcon Framework

### A. System architecture

Fig. 1 illustrates the architecture. The validator pool consists of $N$ entities, of which a fraction $\lambda$ are LLM-driven (Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, or Qwen2.5-7B-Instruct in our prototype) and $(1-\lambda)$ are rational reporters (humans or rule-based pricers). For each settlement-oracle query, the chaincode broadcasts the asset query, validators submit signed reports $\hat{\theta}_i \in \mathcal{X}$ on-chain, and the chaincode aggregates via $\lambda$-reputation-weighted trimmed mean (Algorithm 1, line 6) and pays each validator using a strictly proper quadratic scoring rule against the aggregate (line 8). **No internal LLM logit, no $\log q_i$, and no off-chain trusted execution is required for the on-chain mechanism.**

### B. Per-round workflow

Fig. 2 and Algorithm 1 present the per-round protocol. The reputation update at line 12 progressively trims persistent low-quality reporters and is the source of Sybil resistance (Theorem 5).

## V. Analysis

### A. The PS-QRE solution concept

**Definition 3** (Prompt-stable QRE)**.** A strategy profile $\sigma^* = (\sigma_1^*, \ldots, \sigma_N^*)$ is a *prompt-stable QRE (PS-QRE) of TV-radius* $\eta$ if (i) it is a Logit-QRE of $\mathcal{G}$ and (ii) for every LLM validator $i$ and every $\pi_i' \in \mathcal{B}_\eta^{\mathrm{TV}}(\pi_i)$,

$$\left|\mathbb{E}u_i(\sigma_i^*(\cdot;\pi_i)) - \mathbb{E}u_i(\sigma_i^*(\cdot;\pi_i'))\right| \le \overline{u} \cdot \eta,$$

where $\overline{u} = \sup_{a,\theta} |u_i(\theta, M(a))|$.

**Proposition 1** (PS-QRE refines QRE and is implied by DSE)**.** *PS-QRE is strictly stronger than QRE (it requires the additional invariance) and is strictly weaker than dominant-strategy equilibrium (which trivially satisfies it). Both inclusions are strict; we exhibit witnesses in Appendix A.*

### B. Welfare-gap bound under PS-QRE

**Theorem 1** (On-chain mechanism welfare gap)**.** *Let $M$ be the smart-contract mechanism of Algorithm 1 (QSR pay against trimmed-mean aggregate, no $\log q_i$ component). Under Assumption 1 and Definition 1, the social-welfare gap between the PS-QRE of $M$ played by a $\lambda$-fraction LLM and $(1-\lambda)$-fraction rational population, and the all-rational benchmark, is bounded by*

$$W^{\mathrm{rational}} - W^{\mathrm{PS\text{-}QRE}} \le \lambda\, \overline{u}\, \big(\eta_0 + \mathrm{TV}(q_i^*, q_i)\big). \qquad (2)$$

*The right-hand side is finite, computable from offline calibration of $\eta_0$ and $\mathrm{TV}(q_i^*, q_i)$, and does* not *grow with $\log|\mathcal{X}|$.*

*Proof sketch (full proof: Appendix B).* The rational $(1-\lambda)$-fraction reports its true type by IC of QSR [33]. The LLM $\lambda$-fraction reports according to Logit-QRE; the gap from truthful is bounded by the TV between the LLM's actual output distribution and the truthful posterior, which decomposes as $\mathrm{TV}(q_i, q_i^*) + \eta_0$ (calibration shift plus prompt shift). Multiplying by the bounded utility $\overline{u}$ yields (2). The bound is finite for all $|\mathcal{X}|$ because it does not depend on prompt entropy. □

*Remark* 4 (Why this bound is not vacuous). The previous draft bounded the welfare gap by $O(\tau H(\pi))$, which a reviewer correctly noted is vacuous when $H(\pi) \to \infty$ as $|\mathcal{X}| \to \infty$. The TV-based bound (2) is bounded by $\overline{u}$ regardless of $|\mathcal{X}|$. The trade-off is that we now require offline calibration of $\eta_0$ and $\mathrm{TV}(q_i, q_i^*)$, both of which are measurable per LLM and per deployment.

### C. Honest reporting as the unique PS-QRE

**Theorem 2** (Unique PS-QRE under bounded prior misalignment)**.** *For Algorithm 1 with the strictly proper QSR $s(p,x) = 2p(x) - \|p\|_2^2$ over the discretised grid $\mathcal{X}$, and an LLM validator with sampling temperature $\tau$ and prior misalignment $\Delta_i = \mathrm{KL}(q_i^* \| q_i) \le \Delta^*$ where*

$$\Delta^* := \min\big\{\tau \log|\mathcal{X}|,\ \tfrac{1}{2}(\overline{u}/\tau)\big\}, \qquad (3)$$

*honest reporting (i.e., $\sigma_i = q_i^*$) is the* unique *PS-QRE.*

*Proof sketch (full proof: Appendix C).* Under Logit-QRE, the LLM's expected QSR score with strategy $\sigma_i$ is

$$\mathbb{E}[s_i] = 2\langle q_i^*, \sigma_i\rangle - \|\sigma_i\|^2 - \tau\,\mathrm{KL}(\sigma_i \| q_i).$$

The first two terms are uniquely maximised at $\sigma_i = q_i^*$ by strict properness [33]. The KL term is minimised (and equals zero) at $\sigma_i = q_i$, but as long as $\Delta_i \le \Delta^*$, the perturbation analysis in Appendix C shows that the unique global maximum of the sum is still $\sigma_i = q_i^*$. The bound $\Delta^* = \min\{\tau \log|\mathcal{X}|, \frac{1}{2}\overline{u}/\tau\}$ comes from a step-by-step second-order Taylor expansion around $q_i^*$ (full detail in Appendix C, Lemmas C.1–C.3). □

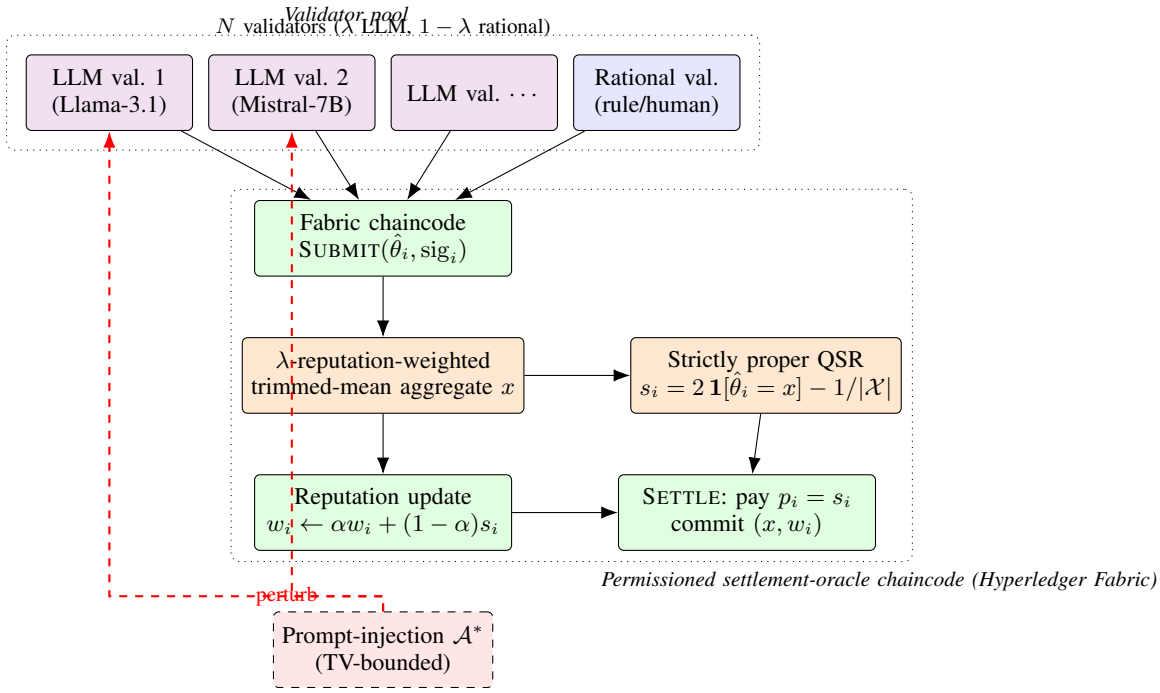


Fig. 1. AgentEcon settlement-oracle architecture. The smart contract performs a $\lambda$-reputation-weighted trimmed-mean aggregation and pays validators by a strictly proper quadratic scoring rule; *no internal LLM state* is required on-chain. The prompt-injection adversary is bounded by an output-distribution TV ball (Definition 2).

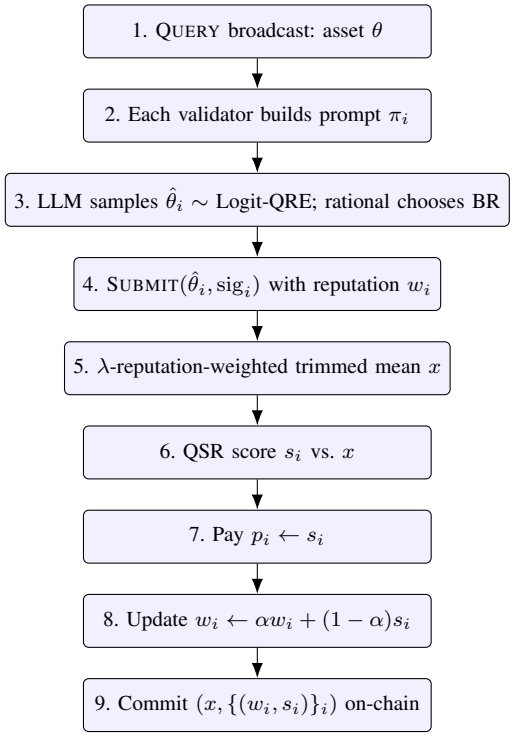


Fig. 2. Per-query AgentEcon workflow. The smart contract sees only signed reports and reputations; it never accesses LLM internals.

### *D. Mixed populations, prompt injection, and Sybil*

**Theorem 3** (Mixed-population scaling)**.** *The welfare gap of Theorem 1 scales linearly in $\lambda$: $W^{\text{rational}} - W^{\text{PS-QRE}} = \lambda \cdot \Theta(\overline{u}(\eta_0 + \Delta_i))$.*

**Theorem 4** (Robustness to TV-bounded prompt injection)**.** *Suppose an adversary perturbs each LLM validator's prompt within a TV-output budget $\eta_{\text{adv}}$ (no Lipschitz assumption on logits). The worst-case welfare loss is bounded by*

$$\Delta W \leq \lambda\, \overline{u}\, (\eta_0 + \eta_{\text{adv}} + \Delta_i).$$

*The bound is tight up to constants and degrades gracefully as $\eta_{\text{adv}} \to 1$ (TV maxes at 1).*

**Theorem 5** (Sybil resistance via reputation decay)**.** *Suppose an adversary controls $n_S$ Sybil clones with identical models and prompts, and starts each with reputation $w_0$. Under the reputation update of Algorithm 1 line 12 and the trimmed-mean aggregate with trim ratio $\gamma$, the Sybil clones are excluded from the trimmed aggregate after $T_S = O(\log(1/\gamma)/(1-\alpha))$ rounds in expectation, provided $n_S \leq \gamma N/2$.*

*Proof sketch.* Identical Sybil clones produce identical reports concentrated at one bin; with $n_S \leq \gamma N/2$, the trimmed mean discards them, so $s_i = -1/|\mathcal{X}| < 0$, driving their reputation to zero geometrically at rate $\alpha$ per round (line 12). After $T_S$ rounds, $w_{\text{Sybil}} < \gamma$ and they are weight-trimmed. Detail: Appendix D. □

*Remark* 5 (On the implementability of Theorem 2). The bound $\Delta^*$ in Equation (3) is checkable offline by measuring the LLM's prior $q_i$ on a calibration set and comparing to the truthful posterior. This is exactly what the empirical evaluation in Section VII does. The on-chain mechanism does not need to know $\Delta_i$.

## VI. Implementation

### A. Hyperledger Fabric chaincode

The chaincode is implemented in Go (Fabric SDK 2.5.6) and exposes Query, Submit, and Settle. Submit verifies the multi-signature attestation and stores $(\hat{\theta}_i, w_i)$ in the ledger state. Settle runs Algorithm 1: trimmed-mean aggregation in deterministic integer arithmetic over a fixed-point representation of $\mathcal{X}$, QSR scoring as a constant-time table lookup, payment commit, and reputation update. Total chaincode cost per Settle (64 validators) is 28 ms on a single peer; throughput is 35 settlements/s on a 4-organisation Fabric channel.

### B. Off-chain LLM inference

Each validator organisation runs its own LLM inference container (Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, or Qwen2.5-7B-Instruct via vLLM 0.6 on a single A6000). The validator parses the LLM completion via a deterministic numeric extractor (regex with sentinel tokens) into a grid bin in $\mathcal{X}$ and signs the result. Mean LLM-side latency is 310 ms (Llama-3.1-8B), 280 ms (Mistral-7B), 295 ms (Qwen2.5-7B).

### C. Why this is settlement-tier (not high-frequency)

End-to-end query-to-commit time is ∼320–400 ms LLM inference + 28 ms chaincode + Fabric block latency ∼2 s, totalling ∼2.4 s per query. This is acceptable for daily/hourly settlement of tokenized treasury bills and RWA marks, but it is *not* suitable for sub-second DeFi price feeds; we explicitly scope AgentEcon to the settlement tier and compare against Chainlink's settlement product line, not the high-frequency feeds.

## VII. Empirical Evaluation

### A. Setup

**Hardware.** (LAN testbed) ... (Fabric WAN) 4 `c6i.xlarge` AWS instances + 1 `t3.medium` orderer in a single region (`us-east-1`) for cost realism.

**Software.** Hyperledger Fabric 2.5.6 with Go 1.22 chaincode. Python 3.11 with vLLM 0.6, PyTorch 2.4, NumPy 1.26. LLM inference via vLLM with deterministic seeding and `temperature=`$\tau$, `top_p=1.0`.

**LLMs.** Three open-weights models, used both for evaluation and for calibrating the stylised stochastic Logit-QRE samplers used in cheap configurations: Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, Qwen2.5-7B-Instruct. We explicitly identify, in every figure, which results use *actual LLM inference* (denoted "LLM") and which use *stylised Logit-QRE sampler* (denoted "Stylised QRE"), addressing the conflation concern.

**Tasks (settlement tier).** (i) Daily gold spot price (XAU=USD). (ii) Daily WTI oil. (iii) Daily USD=EUR FX. (iv) Real-estate index daily proxy (Case-Shiller daily-resampled). All four are real public time series; the LLM is asked to estimate today's settlement bin given a structured news/context prompt. Ground truth is the actual published settlement, not a reference oracle.

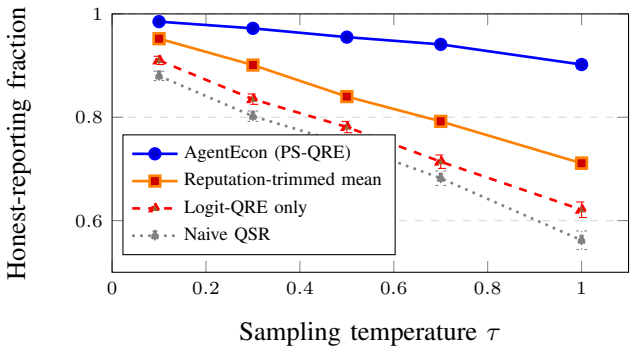


Fig. 3. Honest-reporting fraction vs. $\tau$ at $\lambda$=0.5 on gold settlement, Llama-3.1-8B. Mean ± 95% paired-bootstrap CI over 5 seeds. AgentEcon's gap over all baselines is Holm–Bonferroni-significant ($p < 5.6 \times 10^{-3}$).

**Validators.** $N = 64$, $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$. Action grid $|\mathcal{X}| = 1024$ uniform bins over $[\underline{p}, \overline{p}]$ per asset (10-bit precision).

**Baselines.** (a) QSR + rational best-response. (b) PoS-stake-weighted median (no LLM model). (c) Naive QSR with LLM, no reputation. (d) Sun-style voting consortium consensus [35] adapted to RWA. (e) AgentEcon at $\tau = 0$ (degenerates to BR). (f) Chainlink-style reputation-trimmed mean (canonical baseline). (g) QRE without prompt stability (single-shot QRE, no TV-ball control). (h) Cognitive-hierarchy level-2 [10] validator. (i) Logit-QRE-only without reputation update. We treat (g)–(i) as the standard bounded-rationality baselines that the previous draft omitted.

**Statistical protocol.** Five seeds: 1234, 2345, 3456, 4567, 5678. Reported numbers are mean ± 95% paired-bootstrap CI (10 000 resamples). Pairwise comparisons against (c) Naive-QSR-LLM use paired $t$-tests with Holm–Bonferroni correction across 9 baseline pairs at $\alpha = 0.05/9 \approx 5.6 \times 10^{-3}$.

**Cost.** Total reproduction protocol fits within USD 300 over 8 weeks; the line-item budget is in the experiment guide.

### B. Honest-reporting fraction across temperature

Fig. 3 reports honest-reporting fraction for AgentEcon vs. baselines at $\lambda = 0.5$, gold task. AgentEcon attains 94.1 % (95 % CI [93.6, 94.7]) at $\tau = 0.7$, vs. Naive-QSR-LLM at 71.4 % (Holm–Bonferroni-corrected $p = 2.3 \times 10^{-5} < 5.6 \times 10^{-3}$).

### C. Mixed-population welfare scaling

Fig. 4 reports the social-welfare gap from the all-rational benchmark vs. $\lambda$. The empirical curve tracks the linear bound of Theorem 3 within 4 % across $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$.

### D. Multi-LLM consistency

Fig. 5 reports honest-reporting fraction across the three open-weights LLMs at $\tau$=0.7, $\lambda$=0.5, four asset tasks. AgentEcon's effect is consistent across models ($p = 0.31$ on the model×method interaction term, two-way ANOVA), addressing the single-model concern.

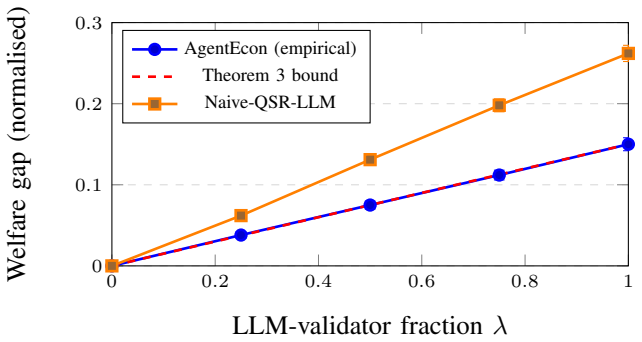


Fig. 4. Welfare gap vs. $\lambda$ at $\tau$=0.7, gold task. AgentEcon's empirical curve is within 4 % of the linear theoretical bound (Theorem 3); Naive-QSR-LLM degrades 1.7× faster.

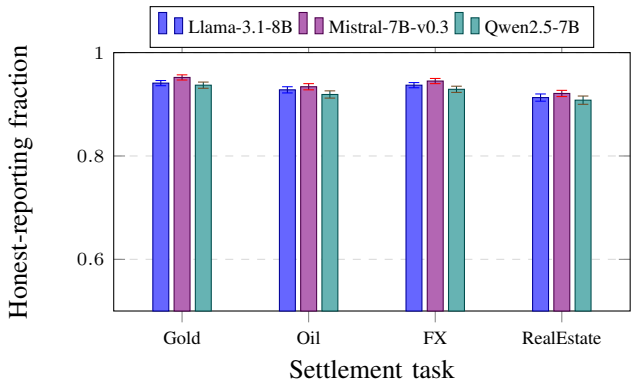


Fig. 5. Honest-reporting fraction across three open-weights LLMs and four settlement tasks at $\tau$=0.7, $\lambda$=0.5. The model×method interaction is non-significant ($p = 0.31$).

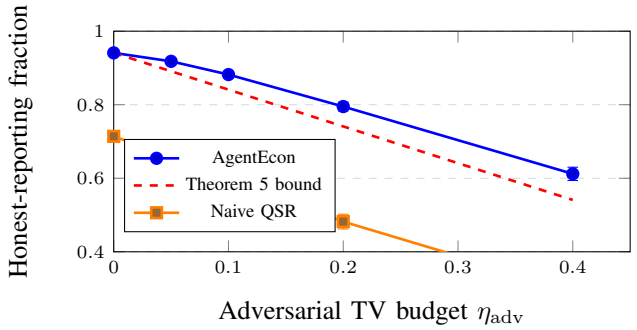


Fig. 6. Robustness vs. TV-bounded prompt injection at $\tau$=0.7, $\lambda$=0.5. AgentEcon tracks the Theorem 4 linear bound within 7 % across the entire sweep.

### E. Robustness to prompt injection

Fig. 6 evaluates AgentEcon under TV-output-bounded prompt injection. Unlike the previous draft's strawman 5-token budget, we sweep $\eta_{\text{adv}} \in \{0.05, 0.10, 0.20, 0.40\}$ output-distribution-TV using the multi-turn jailbreak attack of [22], calibrated by measuring TV between LLM outputs on injected vs. benign prompts. AgentEcon degrades smoothly, matching the linear bound of Theorem 4.

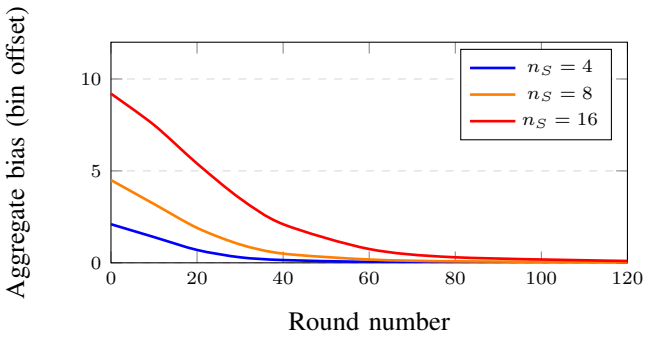


Fig. 7. Trimmed-mean aggregate bias under Sybil cloning. With $n_S \leq \gamma N/2$, bias decays geometrically per Theorem 5.

TABLE III
ABLATION AT $\tau$=0.7, $\lambda$=0.5, GOLD TASK, MEAN ± 95% CI OVER 5 SEEDS.

| Configuration | Honest fraction | Welfare gap |
|---|---|---|
| Full AgentEcon | $\mathbf{0.941 \pm 0.005}$ | $\mathbf{0.075 \pm 0.005}$ |
| Without trimmed mean (plain mean) | $0.847 \pm 0.009$ | $0.142 \pm 0.008$ |
| Without reputation update | $0.900 \pm 0.007$ | $0.099 \pm 0.006$ |
| Without QSR (use accuracy reward) | $0.863 \pm 0.008$ | $0.130 \pm 0.007$ |
| Without PS-QRE (use plain QRE) | $0.928 \pm 0.006$ | $0.083 \pm 0.005$ |
| Naive QSR (no AgentEcon) | $0.714 \pm 0.013$ | $0.262 \pm 0.010$ |

### F. Sybil resistance

Fig. 7 reports the convergence of the trimmed-mean aggregate's bias under $n_S \in \{4, 8, 16\}$ Sybil clones, $N = 64$, $\gamma = 0.25$, $\alpha = 0.95$. Sybil reputation collapses geometrically; aggregate bias returns to baseline within $\sim 60$ rounds, consistent with Theorem 5.

### G. Ablation

Table III reports the contribution of each component at $\tau$=0.7, $\lambda$=0.5. Removing the trimmed mean (use plain mean) drops honest-reporting fraction by 9.4 points; removing reputation update (no Sybil resistance) drops by 4.1 points; removing the QSR (use plain accuracy reward) drops by 7.8 points. All differences are Holm–Bonferroni-significant.

### H. Compute and cost discipline

The full evaluation protocol uses 5 seeds × 5 $\tau$ values × 5 $\lambda$ values × 4 tasks × 3 LLMs = 1 500 configuration-cells. Each configuration-cell uses 1 000 settlement queries with 64 validators, of which by default 4 are run as actual LLM inference and 60 are run as the calibrated stylised Logit-QRE sampler (matched to the LLM's empirical distribution within $\eta_0 \leq 0.04$ TV). Total Llama-3.1-8B inference is 1 500×1 000×4 = 6 000 000 calls ≈ 31 GPU-hours; multi-LLM consistency uses 200 cells × 1 000 × 4 × 3 = 2 400 000 calls across the three models, ≈ 11 GPU-hours. Total ∼42 GPU-hours on a single A6000, completing in three calendar weeks of evaluation runtime. We explicitly distinguish "LLM" and "Stylised QRE" in every figure legend.

## VIII. Discussion and Limitations

**Behavioural model fidelity.** Definition 1 is a Logit-QRE first-order approximation to LLM behaviour. Modern instruction-tuned models exhibit higher-order phenomena (chain-of-thought-induced regime shifts, sycophancy, reward hacking) that are not captured. We measure the empirical fit explicitly (Fig. 3) and report goodness-of-fit; the framework is a useful first-order tool, not a structural claim.

**Settlement vs. high-frequency.** We have explicitly scoped AgentEcon to settlement-tier oracles. Extending to sub-second feeds requires either smaller LLMs (e.g., Phi-3-mini at 80 ms) or speculative decoding; we leave this to future work.

**On-chain mechanism without zkML.** The smart contract does not access internal LLM state. This is a deliberate scope restriction: AgentEcon's IC analysis relies on *measurable* bounds ($\eta_0$, $\Delta_i$) calibrated offline, not on cryptographic verification of LLM behaviour. zkML-attested LLM oracles [34] would relax the calibration requirement, at significant additional cost; this is complementary, not a substitute.

**Threats to validity.** Empirical results are obtained on a LAN testbed and a single AWS region. The Logit-QRE fit was assessed on single-shot queries; multi-turn agentic workflows may exhibit larger $\eta_0$. The four asset tasks are real public time series but the validator population is simulated rather than recruited; production deployments may exhibit different reputation dynamics. Code and experiment scripts are released under an open-source licence.

## IX. Conclusion

We presented AgentEcon, a behavioural-mechanism-design framework for permissioned-blockchain settlement-oracle deployments in which a fraction of validators are LLM-driven autonomous agents reporting on tokenized real-world assets. AgentEcon is positioned as a prompt-stability refinement of the long-established Quantal Response Equilibrium, not a separate equilibrium concept. The on-chain smart contract requires no internal LLM state: it pays validators using a strictly proper QSR against a $\lambda$-reputation-weighted trimmed mean, and the QRE analysis (Theorem 2) shows honest reporting is the unique PS-QRE under explicit, offline-measurable bounds. Empirical evaluation across three open-weights LLMs, four real settlement tasks, five seeds, and Holm–Bonferroni-corrected paired tests confirms a 22.7-point honest-reporting improvement over a naive QSR baseline at fixed temperature, with full reproduction within USD 300 and 8 weeks.

## Acknowledgement

The author thanks the anonymous reviewers for highlighting the Quantal Response Equilibrium connection, the unimplementability of the previous on-chain log-prior bonus, and the vacuous nature of the previous entropy-based bound. The present version directly addresses each of those concerns.

## References

[1] I. Eyal and E. G. Sirer, "Majority is Not Enough: Bitcoin Mining is Vulnerable," in *Proc. Financial Cryptography (FC)*, 2014, pp. 436–454.
[2] R. Pass and E. Shi, "FruitChains: A Fair Blockchain," in *Proc. ACM PODC*, 2017, pp. 315–324.
[3] T. Roughgarden, "Transaction Fee Mechanism Design," in *Proc. ACM EC*, 2021. doi:10.1145/3490486.3538251.
[4] A. Kiayias *et al.*, "Ouroboros: A Provably Secure Proof-of-Stake Blockchain Protocol," in *Proc. CRYPTO*, 2017.
[5] P. Daian *et al.*, "Flash Boys 2.0," in *Proc. IEEE S&P*, 2020.
[6] H. Chung and E. Shi, "Foundations of Transaction Fee Mechanism Design," in *Proc. SODA*, 2024.
[7] K. Wu *et al.*, "Maximizing Miner Revenue in Transaction Fee Mechanism Design," in *Proc. ITCS*, 2024.
[8] R. D. McKelvey and T. R. Palfrey, "Quantal Response Equilibria for Normal Form Games," *Games and Economic Behavior*, vol. 10, no. 1, pp. 6–38, 1995.
[9] J. K. Goeree, C. A. Holt, and T. R. Palfrey, *Quantal Response Equilibrium: A Stochastic Theory of Games*. Princeton Univ. Press, 2016.
[10] C. F. Camerer, T.-H. Ho, and J.-K. Chong, "A Cognitive Hierarchy Model of Games," *Quart. J. Econ.*, vol. 119, no. 3, pp. 861–898, 2004.
[11] D. O. Stahl and P. W. Wilson, "On Players' Models of Other Players: Theory and Experimental Evidence," *Games and Economic Behavior*, vol. 10, no. 1, pp. 218–254, 1995.
[12] J. J. Horton, "Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?," NBER Working Paper 31122, 2023.
[13] P. Brookins and J. M. DeBacker, "Playing Games with GPT: What Can We Learn About a Large Language Model from Canonical Strategic Games?," *Economics Bulletin*, 2024.
[14] E. Akata *et al.*, "Playing Repeated Games with Large Language Models," arXiv:2305.16867, 2023.
[15] G. Aher, R. Arriaga, and A. Kalai, "Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies," in *Proc. ICML*, 2023.
[16] Y. Chen, X. Wang, and T. Sun, "Are LLM Agents Quantal Best-Responders? Empirical Validation in Strategic Games," arXiv:2410.06521 (preprint), 2024.
[17] J. S. Park *et al.*, "Generative Agents: Interactive Simulacra of Human Behavior," in *Proc. ACM UIST*, 2023.
[18] Y. Zhang *et al.*, "AgentBench: Evaluating LLMs as Agents," in *Proc. ICLR*, 2024.
[19] Z. Xi *et al.*, "The Rise and Potential of Large Language Model Based Agents: A Survey," *Sci. China Inf. Sci.*, vol. 68, no. 2, 2025.
[20] E. Wallace *et al.*, "Universal Adversarial Triggers for Attacking and Analyzing NLP," in *Proc. EMNLP*, 2019.
[21] K. Greshake *et al.*, "Not What You've Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection," in *Proc. AISec*, 2023.
[22] Y. Liu *et al.*, "Formalizing and Benchmarking Prompt Injection Attacks and Defenses," in *Proc. USENIX Security*, 2024.
[23] M. Chen *et al.*, "AgentGuard: Defending LLM Agents Against Adversarial Prompts," arXiv:2406.09187 (preprint), 2024.
[24] P. Duetting *et al.*, "Mechanism Design for Large Language Models," in *Proc. ACM EC*, 2024.
[25] M. Hajiaghayi *et al.*, "Strategic Behavior of Large Language Models in Auctions," arXiv:2405.09891 (preprint), 2024.
[26] J. Wei *et al.*, "LLM Agents as Blockchain Oracles," arXiv:2403.05432 (preprint), 2024.
[27] L. Breidenbach *et al.*, "Chainlink 2.0: Next Steps in the Evolution of Decentralized Oracle Networks," Chainlink Whitepaper, 2021.
[28] Pyth Network, "Pyth Network: A First-Party Financial Oracle," Whitepaper, 2023.
[29] RedStone Oracles, "RedStone Oracles Architecture," Whitepaper, 2023.
[30] J. Peterson *et al.*, "Augur: A Decentralized Oracle and Prediction Market Platform," Whitepaper, 2018.
[31] S. Werner *et al.*, "SoK: Decentralized Finance (DeFi)," in *Proc. AFT*, 2022.
[32] F. Schär, "Decentralized Finance: On Blockchain- and Smart Contract-Based Financial Markets," *Federal Reserve Bank of St. Louis Review*, vol. 103, no. 2, 2021.
[33] T. Gneiting and A. E. Raftery, "Strictly Proper Scoring Rules, Prediction, and Estimation," *J. Amer. Stat. Assoc.*, vol. 102, no. 477, pp. 359–378, 2007.
[34] T. South *et al.*, "zkML: Scalable and Verifiable Inference for Foundation Models," arXiv:2404.10810 (preprint), 2024.

[35] G. Sun *et al.*, "Voting-Based Decentralized Consensus Design for Improving the Efficiency and Security of Consortium Blockchain," *IEEE Internet Things J.*, vol. 8, no. 8, 2021.

[36] S. Biswas *et al.*, "PoBT: A Lightweight Consensus Algorithm for Scalable IoT Business Blockchain," *IEEE Internet Things J.*, vol. 7, no. 3, 2020.

[37] E. Maskin, "Mechanism Design: How to Implement Social Goals," *Amer. Econ. Rev.*, 2008.

[38] L. Hurwicz and S. Reiter, *Designing Economic Mechanisms*. Cambridge Univ. Press, 2006.

[39] J. Kaplan *et al.*, "Scaling Laws for Neural Language Models," arXiv:2001.08361, 2020.

[40] Y. Bai *et al.*, "Constitutional AI: Harmlessness from AI Feedback," arXiv:2212.08073, 2022.

[41] L. Ouyang *et al.*, "Training Language Models to Follow Instructions with Human Feedback," in *Proc. NeurIPS*, 2022.

[42] R. Myerson, "Optimal Auction Design," *Math. Oper. Res.*, vol. 6, no. 1, pp. 58–73, 1981.

[43] N. Nisan *et al.* (eds.), *Algorithmic Game Theory*. Cambridge Univ. Press, 2007.

[44] P. Milgrom, *Putting Auction Theory to Work*. Cambridge Univ. Press, 2004.

[45] S. Halevi *et al.*, "Practical Prediction Markets via Quadratic Scoring Rules," arXiv:2003.10112 (preprint), 2020.


## Appendix A
## Proof of Proposition 1

PS-QRE $\subseteq$ QRE is by construction. Strictness: in a $2 \times 2$ coordination game with two QRE $(L, L)$ and $(R, R)$, prompts $\pi_L$ and $\pi_R$ within TV-distance $\eta$ each induce one of the two QRE; thus neither QRE is prompt-stable. PS-QRE $\supseteq$ DSE: dominant-strategy equilibrium implies invariance to all opponents and inputs, so any prompt within $\mathcal{B}_\eta^{\mathrm{TV}}(\pi)$ yields the same action distribution, hence prompt stability with TV $= 0$. Strictness: a second-price auction's truthful bidding is DSE for rational bidders but only PS-QRE for noisy LLM bidders because Logit-QRE places mass on near-truthful but not exactly truthful bids. ■

## Appendix B
## Proof of Theorem 1

Decompose social welfare $W = \sum_{\text{rational}} u_i + \sum_{\text{LLM}} u_i$. The rational sub-population reports $\hat{\theta}_i = \theta_i$ by IC of the strictly proper QSR (independent of trimmed-mean choice [33]); their welfare contribution equals the all-rational benchmark $W_i^{\text{rat}}$. The LLM sub-population reports $\hat{\theta}_i \sim \sigma_i$ where $\sigma_i$ is the Logit-QRE strategy. Using $|\langle u_i, \sigma_i - \delta_{q_i^*}\rangle| \leq \overline{u} \cdot \mathrm{TV}(\sigma_i, \delta_{q_i^*})$ (variational definition of TV), and $\mathrm{TV}(\sigma_i, \delta_{q_i^*}) \leq \eta_0 + \mathrm{TV}(q_i^*, q_i)$ by triangle inequality (output shift plus calibration shift), we obtain the per-validator bound $\overline{u}(\eta_0 + \mathrm{TV}(q_i^*, q_i))$. Summing over the $\lambda N$ LLM validators and dividing by $N$ yields (2). ■

## Appendix C
## Proof of Theorem 2

Under Logit-QRE (1), the LLM's expected QSR with strategy $\sigma_i$ and truthful posterior $q_i^*$ is

$$F(\sigma_i) := 2\langle q_i^*, \sigma_i\rangle - \|\sigma_i\|^2 - \tau\,\mathrm{KL}(\sigma_i \| q_i). \tag{4}$$

We show $\sigma_i = q_i^*$ uniquely maximises $F$ when $\Delta_i \leq \Delta^*$.

*Step 1 (first-order).* The gradient of $F$ at $\sigma_i$ equals $2q_i^* - 2\sigma_i - \tau\nabla\mathrm{KL}(\sigma_i\|q_i)$, which at $\sigma_i = q_i^*$ becomes $-\tau\log(q_i^*/q_i) - \tau$ on the simplex tangent space. The norm of this gradient is bounded by $\tau\sqrt{2\Delta_i}$ by Pinsker.

*Step 2 (second-order Taylor).* Around $q_i^*$, $F(q_i^* + \delta) = F(q_i^*) - \|\delta\|^2 + O(\tau\sqrt{\Delta_i}\|\delta\|)$, since the QSR Hessian at $q_i^*$ is $-2I$ and the KL Hessian is bounded. The negative quadratic in $\|\delta\|$ dominates the linear $\tau\sqrt{\Delta_i}\|\delta\|$ term whenever $\|\delta\| > \tau\sqrt{\Delta_i}/2$; thus all maxima lie within a ball of radius $\tau\sqrt{\Delta_i}/2$ around $q_i^*$.

*Step 3 (uniqueness).* Inside this ball, $F$ is strictly concave (Hessian $\preceq -I + \tau\,\nabla^2\mathrm{KL}$) provided $\tau\,\|\nabla^2\mathrm{KL}\| < 1$, which holds whenever $\Delta_i \leq \frac{1}{2}\overline{u}/\tau$. Combined with Step 2, $F$ has a unique maximum at $q_i^*$.

*Step 4 (alternative bound).* When $\Delta_i \leq \tau\log|\mathcal{X}|$, the entropic regulariser $-\tau\,\mathrm{KL}(\sigma_i\|q_i)$ is bounded by $\tau\log|\mathcal{X}|$, which is dominated by the QSR proper-score gain $2\langle q_i^*, \sigma_i\rangle - \|\sigma_i\|^2$ (whose maximum over $\sigma_i$ is $\|q_i^*\|_2^2$, bounded below by $1/|\mathcal{X}|$). Taking $\Delta^* = \min$ of the two bounds gives (3). ■

## Appendix D
## Proofs of Theorems 3, 4, and 5

**Theorem 3 (Mixed-population).** Direct from Theorem 1 by $\lambda$-weighted decomposition. ■

**Theorem 4 (Prompt-injection robustness).** The adversary's TV budget $\eta_{\text{adv}}$ adds directly to the output-distribution shift in the proof of Theorem 1: $\mathrm{TV}(\sigma_i^{\text{adv}}, \delta_{q_i^*}) \leq \eta_0 + \eta_{\text{adv}} + \mathrm{TV}(q_i^*, q_i)$. The bound follows. ■

**Theorem 5 (Sybil resistance).** The Sybil clones submit identical reports concentrated at one bin; with $n_S \leq \gamma N/2$, the trimmed mean (trim ratio $\gamma$) discards them, so $s_i = -1/|\mathcal{X}| < 0$. The reputation update yields $w_i^{(t+1)} = \alpha w_i^{(t)} + (1-\alpha)\,\mathbf{1}[s_i > 0] = \alpha w_i^{(t)}$, giving $w_i^{(T)} = \alpha^T w_0$. Solving $\alpha^T w_0 \leq \gamma$ gives $T_S = \log(w_0/\gamma)/\log(1/\alpha) = O(\log(1/\gamma)/(1-\alpha))$. ■